UTOPIA ELECTRONIC DEVELOPMENT SPECIFICATIONS

Last updated 2/15/25

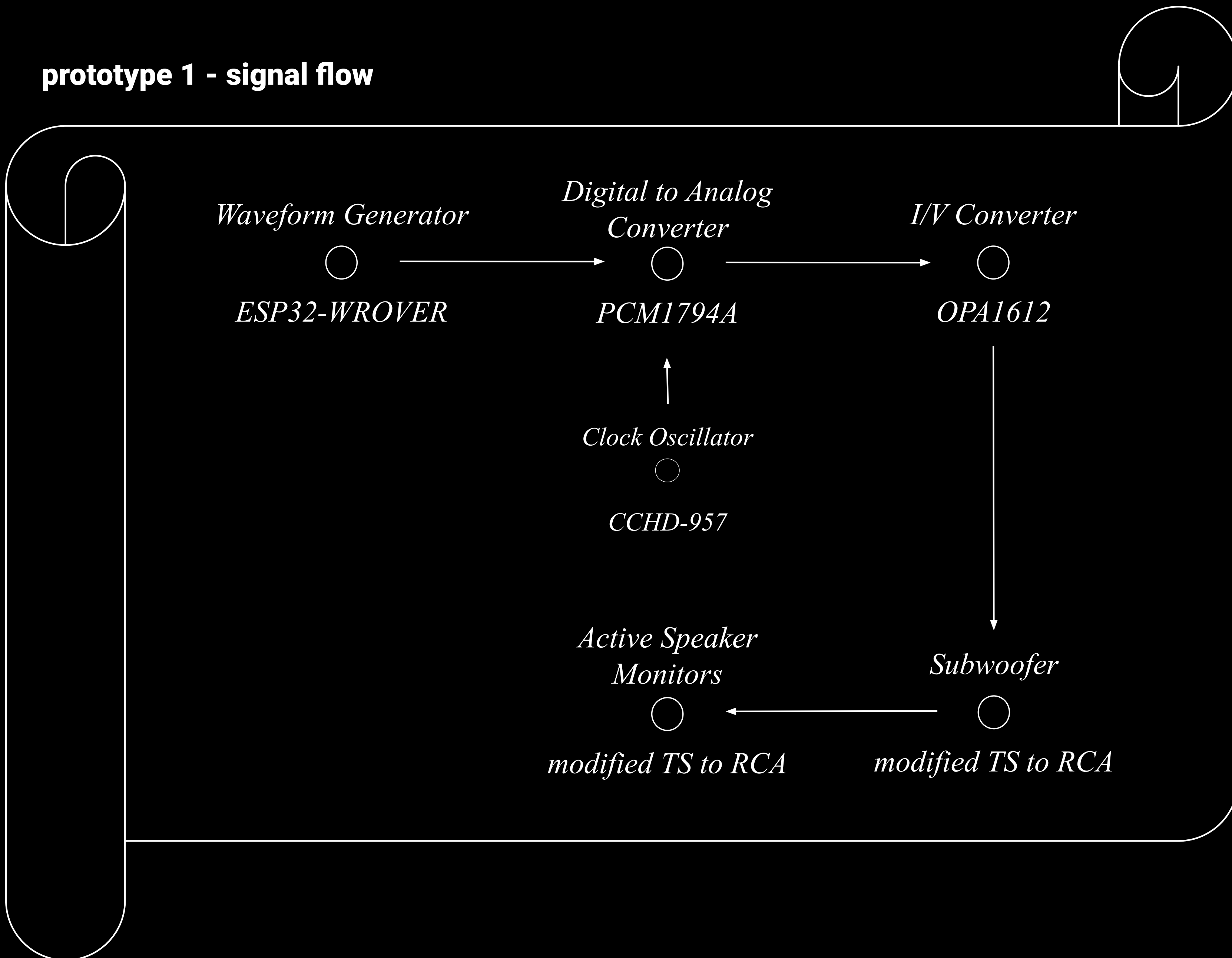
PROJECT OBJECTIVES

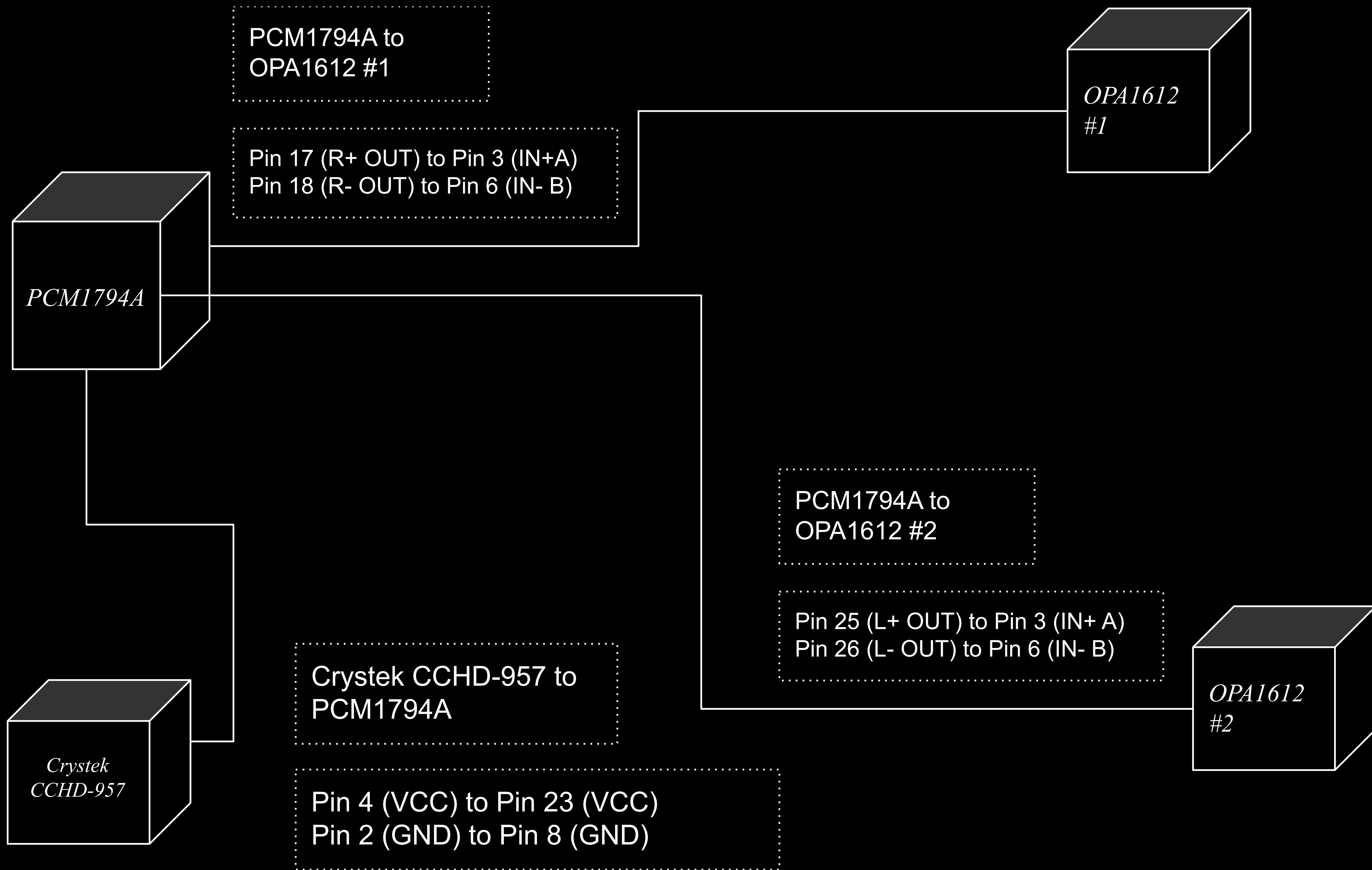
- Design a system of **ESP-32 microcontrollers** programmed to perform **waveform synthesis** and **audio effects** manipulated by **data** from **proximity sensors**.
- Integrate an **oscilloscope visual** for monitoring the performance of sound.
- Integrate **an external sound system** for the hardware synthesizer to output sound.

DEVELOPMENT FLOW

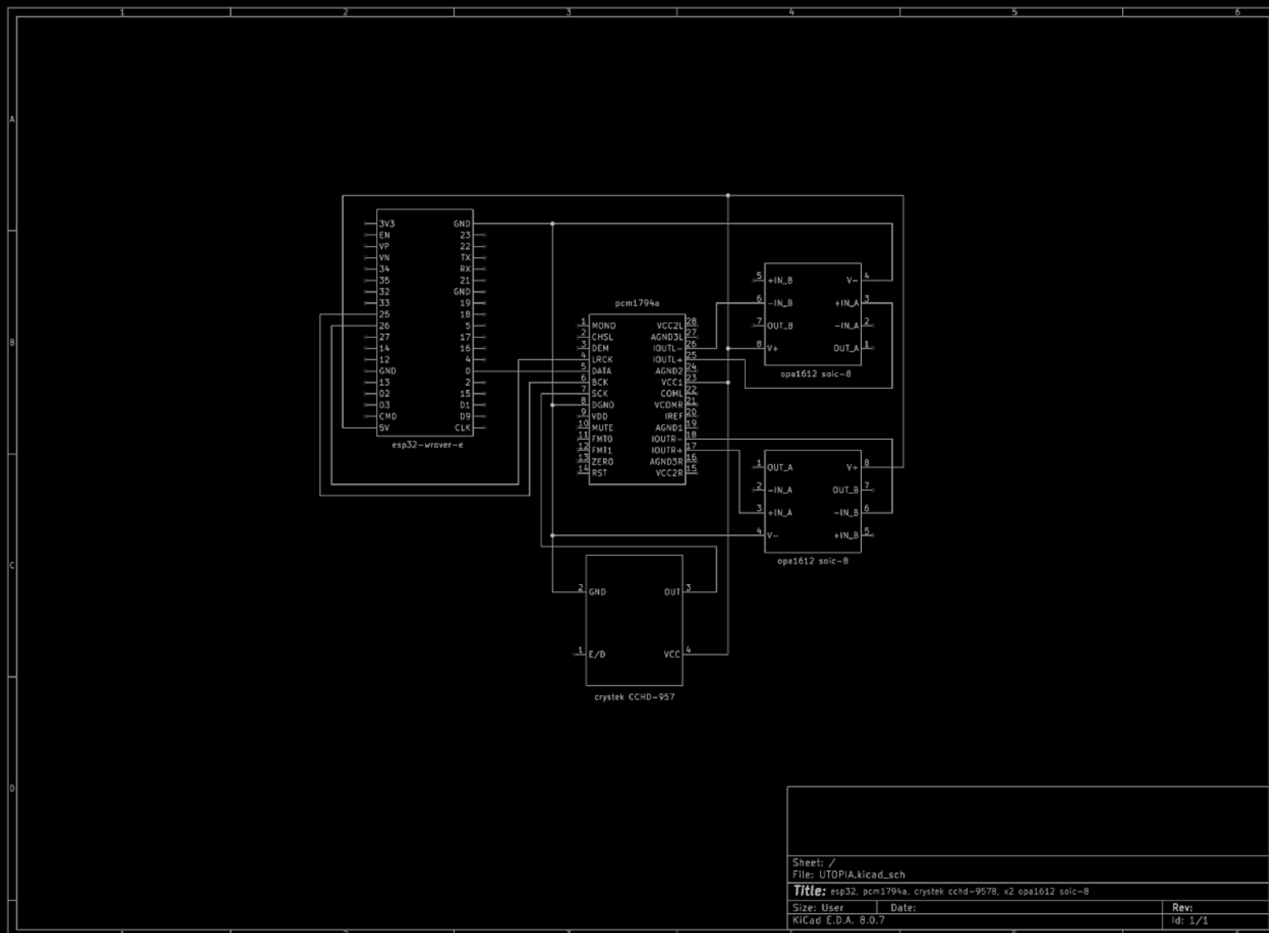
1. internal waveform generation to Speaker output
2. internal audio effect(s)
3. oscillator output
4. microphone input

prototype 1 - signal flow

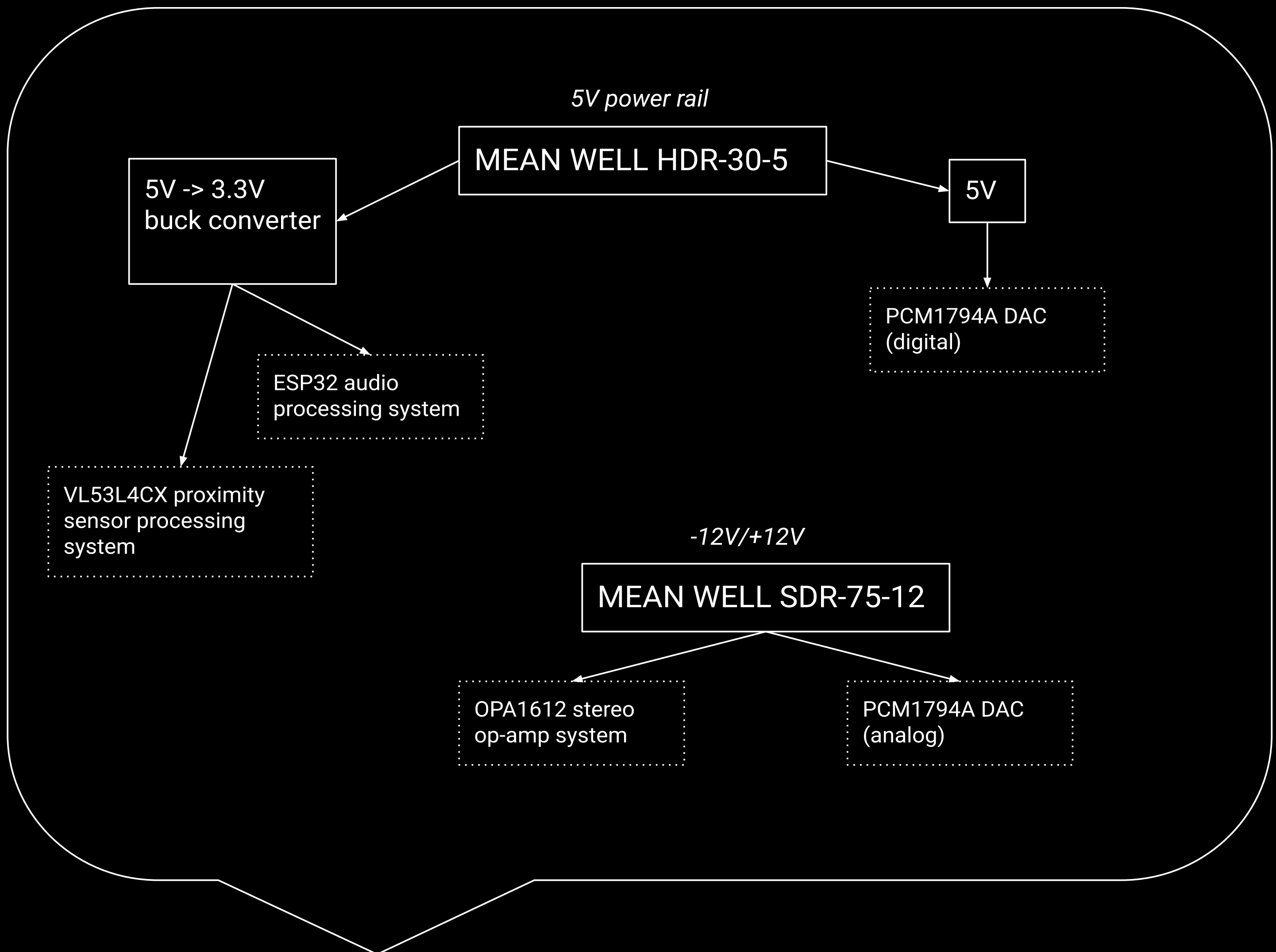




prototype 1 - pinout



prototype 1 - schematic



prototype 1 - power systems

MICROCONTROLLERS: 7 x ESP32-WROVER microcontrollers for waveform generation and audio effect processing.

SENSORS: Input from 11 x VL53L4CX proximity sensors to control audio effect parameters.

DAC: 1 x PCM1794A DAC for digital-to-analog conversion.

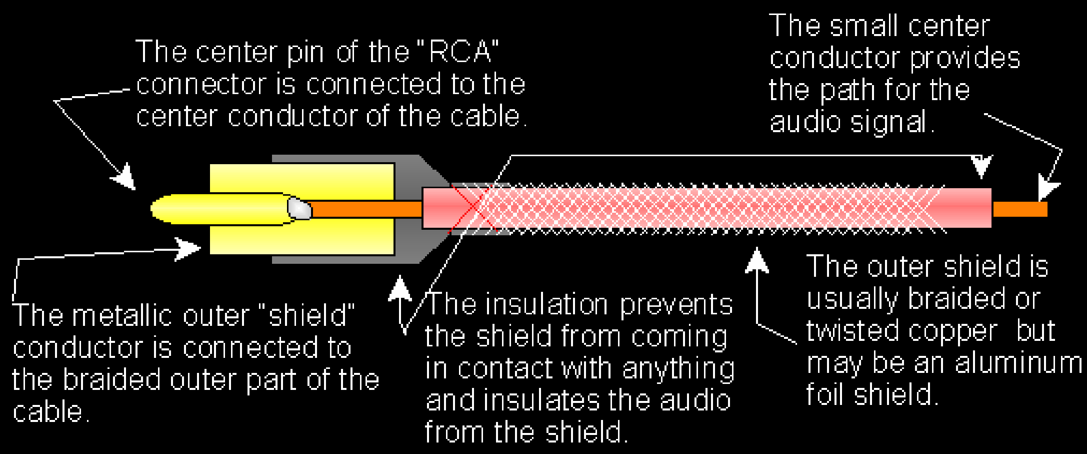
CLOCK: Crystek CCHD-957 clock generator for jitter reduction and synchronization with the DAC.

Signal Processing: 2 x OPA1612 op-amps acting as I/V converters for signal output.

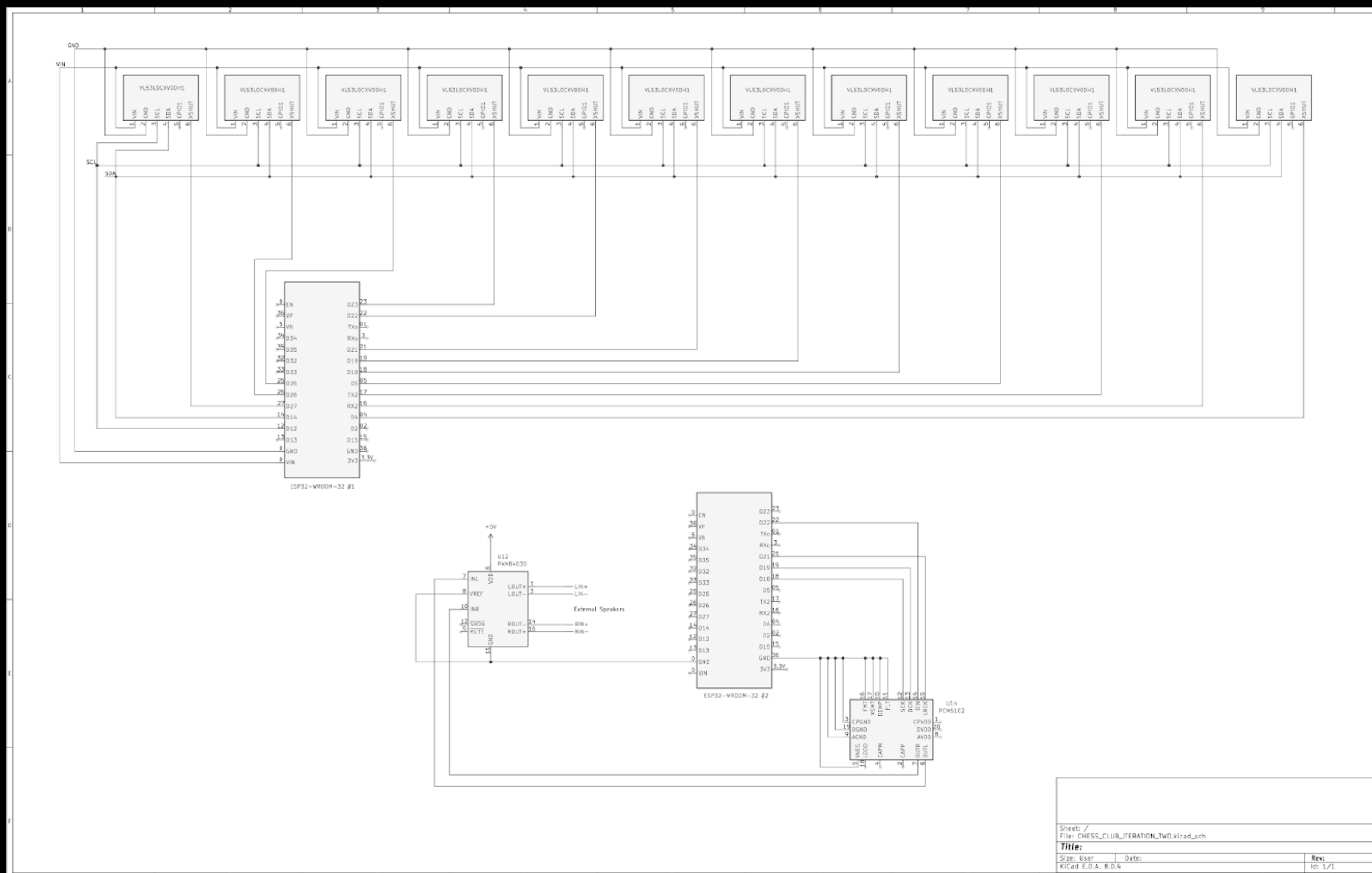
Output: RCA-only subwoofer, outputting to 2 x Yamaha HS5 speakers for final sound reproduction.

Power: 5V MEAN WELL HDR-30-5 to power ESP32s and VL53L4CX sensors via a 5V to 3.3V buck converter. A ±12V dual power MEAN WELL SDR-75-12 to power the OPA1612 op amps and the analog section of the PCM1794A.

Notes: A transformer may be necessary in addition to shielded RCA cables



ARCHIVE



PRELIMINARY SCHEMATICS SUBMITTED FOR PROPOSAL

Preliminary Schematic Overview:
12 VL53LOCX proximity sensors connected to one ESP32.
PCM5102 DAC > ESP32 > **PAM84030** Audio Driver > R/L External Speakers.

Note for modifications in the next version:
2 dedicated ESP32s for processing sensor data
VL53L4CX substitute

PRELIMINARY SCHEMATIC OVERVIEW

** power*

- *5V power supply: MAX98357 amplifier, PCM1794A DAC*
- *5V to 3.3V buck converter*
- *(-12V/+12V) dual power rail*

** inputs*

- *sensors*
    - *VIN to 3.3V*
    - *GND to GND*
    - *SCL to ESP32 SCL*
    - *SDA to ESP32 SDA*
    - *GPIO (none)*
    - *XSHUT to unique GPIO on ESP32*
- *microphone*

** processors*

- *ESP32s*
- *amplifiers*

- *DAC (Digital-to-Analog Converter)*

** outputs*

- *speakers*
- *subwoofer*
- *oled display*